

Welcome everyone to GradieNTAG. This group was first organized by Nathan's two former student, Ahmad and Sharon. The purpose of this reading group/seminar is for students to get more opportunity to present their research or reading.

▽NTAG

The name GradieNTAG is a word play on the gradient: the direction of steepest descent, so we hope that we will reach our goal with the fastest path. It is also an NTAG group for grad students only, so it is also gradie-NTAG.

We specifically did not invite any faculty members so that this will be an environment with relatively less pressure from our supervisors.

Overview

- Moduli space: classification problem, want to classify objects up to equivalence
- Moreover, we want the space of all objects to have structure: family of objects
 - Example: $\text{Gr}(n,k)$, k -dim subspaces of k^n (no equivalence)
 - Example: M_g , proj. smooth curves of genus g /iso

Schemes: (Hartshorne II.1 ~ II.2)

A topological space X w/ a sheaf of rings $\mathcal{O}_X$ on X
satisfying some properties structure sheaf

↳ a locally ringed space, with an affine open covering

DON'T COVER(?)

↳ Sheaf of rings on a topological space:

- On each open $U \subseteq X$, has a ring $\mathcal{O}_X(U)$
 - For each open $U \subseteq V \subseteq X$, has restriction map $\rho_{VU}: \mathcal{O}_X(V) \rightarrow \mathcal{O}_X(U)$
- satisfying some properties

↳ Identity, gluing axioms

Will mainly focus on affine and projective schemes
(over an algebraically closed field). Defined below:

Affine schemes (the object):

Let A be a ring. Have a scheme $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ given by:

- Topological space: $\text{Spec } A$
- Structure sheaf: $\mathcal{O}_{\text{Spec } A}$, where $\mathcal{O}_{\text{Spec } A}(\text{Spec } A) = A$, $\mathcal{O}_{\text{Spec } A}(\underbrace{D_f}) = A_f$.
Open set where $f \neq 0$
 $\{p \in \text{Spec } A: f \notin p\}$

Usually just write $\text{Spec } A$ for the scheme.

Morphisms between affine schemes:

Let $X = \text{Spec } A$, $Y = \text{Spec } B$ be affine schemes.

Prop (II 2.3) Every morphism $f: X \rightarrow Y$

(Def: $f: X \rightarrow Y$ map of spaces
 $f^\#: \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$ map of sheaves on Y)

is given by a homomorphism (of rings)

$$\varphi: B \rightarrow A$$

s.t. $f(p) = \underbrace{\varphi^{-1}(p)}$
is a prime ideal in B

Scheme in general: $(X, \mathcal{O}_X)$ is a scheme if $\exists$ affine open cover
 $\{U_i\}_i$ s.t. $X = \bigcup_{i \in I} U_i$, $\mathcal{O}_X|_{U_i} = \mathcal{O}_{U_i}$ (as sheaves on U_i)
 $U_i = \text{Spec } A_i$

Schemes over some base scheme Y :

Objects: Scheme X w/ a map $X \rightarrow Y$

Morphisms: Given $X_1 \rightarrow Y, X_2 \rightarrow Y$.

A morphism f from $X_1 \rightarrow Y$ to $X_2 \rightarrow Y$ is given by

$$\underbrace{X_1 \xrightarrow{f} X_2}_{\text{morphism of schemes.}} \quad \text{s.t.} \quad \begin{array}{ccc} X_1 & \xrightarrow{f} & X_2 \\ & \searrow & \swarrow \\ & Y & \end{array} \text{ commutes.}$$

• Note ①: Every scheme can be thought of as a scheme over $\text{Spec } \mathbb{Z}$

Since for every ring A , $\exists!$ map $\mathbb{Z} \rightarrow A$
(given by $1 \mapsto 1_A$)

• Note ②: We often think about schemes over a field k .
i.e. over $\text{Spec } k$

We call schemes over k **k -schemes**

Schemes of finite type over k :

Def: $X \rightarrow \text{Spec } k$ is **finite type** if

$\exists$ finite affine open cover $\text{Spec } A_1, \dots, \text{Spec } A_n \subseteq X$ s.t.

A_i is a f.g. k -algebra.

In this case: Can write $A_i = \frac{k[x_1, \dots, x_r]}{I_i}$ as a quotient.

From now on, we assume every scheme is **finite type**
over an algebraically closed field k .

Besides affine schemes, also want to define projective schemes (over k)
(Vakil §4.5, Hartshorne I.2, I.7)

Def: A **graded k -algebra** S is a k -algebra with
a decomposition $S = \bigoplus_{d \geq 0} S_d$ of k -vector spaces such that
 $\forall d, e \geq 0, S_d \cdot S_e \subseteq S_{d+e}$

An element of S_d is called a **homogeneous element of degree d** .

An ideal $a \subseteq S$ is a **homogeneous ideal** if $a = \bigoplus_{d \geq 0} (a \cap S_d)$,
or equivalently, there exists a set of homogeneous generators,
i.e. $\exists s_1, \dots, s_r \in S, s_i \in S_{d_i}$ s.t. $a = (s_1, \dots, s_r)$

E.g. of gr. k -alg.
• $S = k[X_0, \dots, X_n]$
• $S = k[X_0, \dots, X_n]/I$ where $I \subseteq k[X_0, \dots, X_n]$ homogeneous

Projective scheme (Vakil 4.5.7, Hartshorne I.2)

Suppose $S = k[X_0, \dots, X_n]/I$ is a fg. graded k -algebra (I needs to be homog.)

We have a k -scheme $\text{Proj } S$ with an affine cover $\{U_0, \dots, U_n\}$
where $U_i = \text{Spec } k[\frac{X_0}{X_i}, \dots, \frac{X_n}{X_i}]/\bar{I}$ "Open affine patch $X_i \neq 0$ "

For full proj. construction, see references listed above.

e.g. • $\mathbb{P}_k^n := \text{Proj } k[X_0, \dots, X_n]$

• An elliptic plane curve: $\text{Proj}(k[X_0, X_1, X_2]/(X_2^2 X_0 - X_1^3 - X_1 X_0^2))$

(Write x_{y_0} shorthand for $\frac{y_1}{x_0}$) $\hookrightarrow$ Patch $x_0 \neq 0$: $\text{Spec } k[x_{y_0}, x_{y_1}] / (x_{y_0}^2 - x_{y_0}^3 - x_{y_0})$
 $x_1 \neq 0$: $\text{Spec } k[x_{0/y_1}, x_{2/y_1}] / (x_{2/y_1}^2 x_{0/y_1} - 1 - x_{0/y_1}^2)$
 $x_2 \neq 0$: $\text{Spec } k[x_{0/y_2}, x_{y_2}] / (x_{0/y_2} - x_{y_2}^3 - x_{y_2} x_{0/y_2}^2)$

Note: x_{y_0} is degree 0.

Example of morphisms between schemes:

① $A^2 \hookrightarrow P^2$

$$\text{Spec } k[y_1, y_2] \rightarrow \text{Proj } k[x_0, x_1, x_2]$$

$$k[x_0, x_1, x_2] \rightarrow k[y_1, y_2]$$

$$x_0 \mapsto 1$$

$$x_1 \mapsto y_1$$

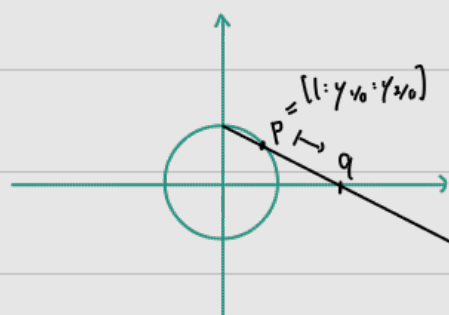
$$x_2 \mapsto y_2$$

② $\text{Proj } k[y_0, y_1, y_2] / \overset{I}{(y_0^2 - y_1^2 - y_2^2)} \rightarrow \text{Proj } k[x_0, x_1]$

$$k[x_0, x_1] \rightarrow k[y_0, y_1, y_2] / I$$

$$x_0 \mapsto y_0 - y_2$$

$$x_1 \mapsto y_1$$



$$\text{slope} = \frac{y_{y_0} - 1}{y_{y_0}}$$

$$\therefore q = [1: \frac{y_{y_0}}{1 - y_{y_0}}] = [1 - y_{y_0}: y_{y_0}] = [y_0 - y_2: y_1]$$

Hilbert polynomial of a projective k -scheme: (Ref. Hartshorne I.7, III.5, Vakil 18.6)

Def Suppose $S = k[x_0, \dots, x_n] / I$ is a graded k -algebra (I has to be homog.)

Then the Hilb. function $HF_S(t)$ of $\text{Proj } S$ (or Hilb. function of S)

is $HF_S(t) := \dim_k S_t$ (degree t piece)

Def-Prop

$\exists$ a polynomial $HP_s(t) \in \mathbb{Q}(t)$ s.t.
 $HP_s(t) = HF_s(t)$ for $t \gg 0$.

e.g. ① Let $S = k[X_0, X_1, X_2]$. Then

t	0	1	2	3	4	5
$\binom{t+2}{2} = HF_s(t)$	1	3	6	10	15	21

$$HP_s(t) = \frac{(t+2)(t+1)}{2} = \frac{1}{2}t^2 + \frac{3}{2}t + 1$$

② Let $S = k[X_0, X_1, X_2] / (X_0^2 - X_1^2 - X_2^2)$

t	0	1	2	3	4	5
$HF_s(t)$	1	3	5	7	9	11

$$HP_s(t) = 2t + 1$$

③ Let $S = k[X_0, X_1, X_2] / (X_2^2 X_0 - X_1^3 - X_0 X_2^2)$

t	0	1	2	3	4	5
$HF_s(t)$	1	3	6	9	12	15

$$HP_s(t) = 3t$$

Exercise:

Calculate $HF_s(t)$
for some t .

Fact: Hilb. poly contains info about many invariants of $\text{Proj } S$.

- $\deg HP_s(t) = \dim \text{Proj } S$.

- Def-Prop The degree of $\text{Proj } S$ is defined to be:

Let $n := \dim \text{Proj } S$

$$\deg \text{Proj } S = n! \cdot [t^n] HP_s(t) \quad \text{Prop: This is an integer}$$

↳ Examples 1, 2, 3 have degrees 1, 2, 3 resp.

↳ Let $f \in k[X_0, \dots, X_r]$ homog. of degree d .

Then $\deg \text{Proj } k[X_0, \dots, X_r]_f = d$.

- Arithmetic genus of $\text{Proj } S = (-1)^{\dim \text{Proj } S} (HP_S(0) - 1)$

Flat morphisms (treat as black box) (Hartshorne III.9 for facts)

We often want a family
 (we treat the morphism $\pi: X \rightarrow Y$ as
 a family parametrized by Y)

$$\begin{array}{c} X \\ \downarrow \pi \\ Y \end{array}$$

where each fiber $\pi^{-1}(y)$ (sometimes written X_y) is an object we want to study.

Want this family to be well-behaved.

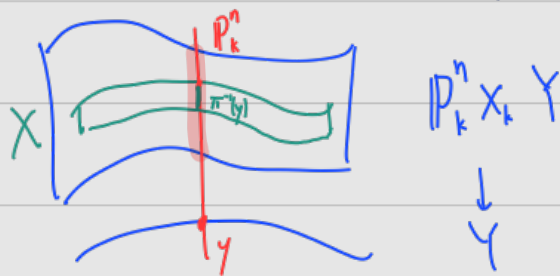
Turns out **flatness** is the "correct" definition.

Facts: Let $\pi: X \rightarrow Y$ be a flat morphism where Y irred. Then:

- (Hart Cor 9.6) Fiber dimension, i.e. $\dim \pi^{-1}(y)$, remains constant.
- If $\dim \pi^{-1}(y) = 0$, then #pts constant.
- (Hart Thm 9.9) Suppose $X \subseteq \mathbb{P}_Y^n := \mathbb{P}_k^n \times_k Y$,

for each point $y \in Y$, consider Hilb. poly of $\pi^{-1}(y) \subseteq \mathbb{P}_k^n$

Then $\pi: X \rightarrow Y$ is flat $\Leftrightarrow$ Hilb. poly independent of y .



- Refer to Hart III.10 for more facts regarding flat/smooth morphisms.